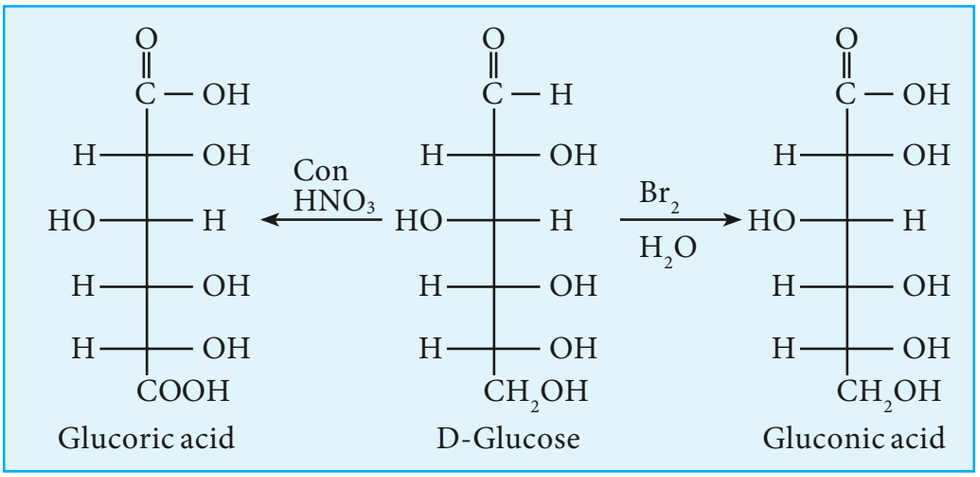
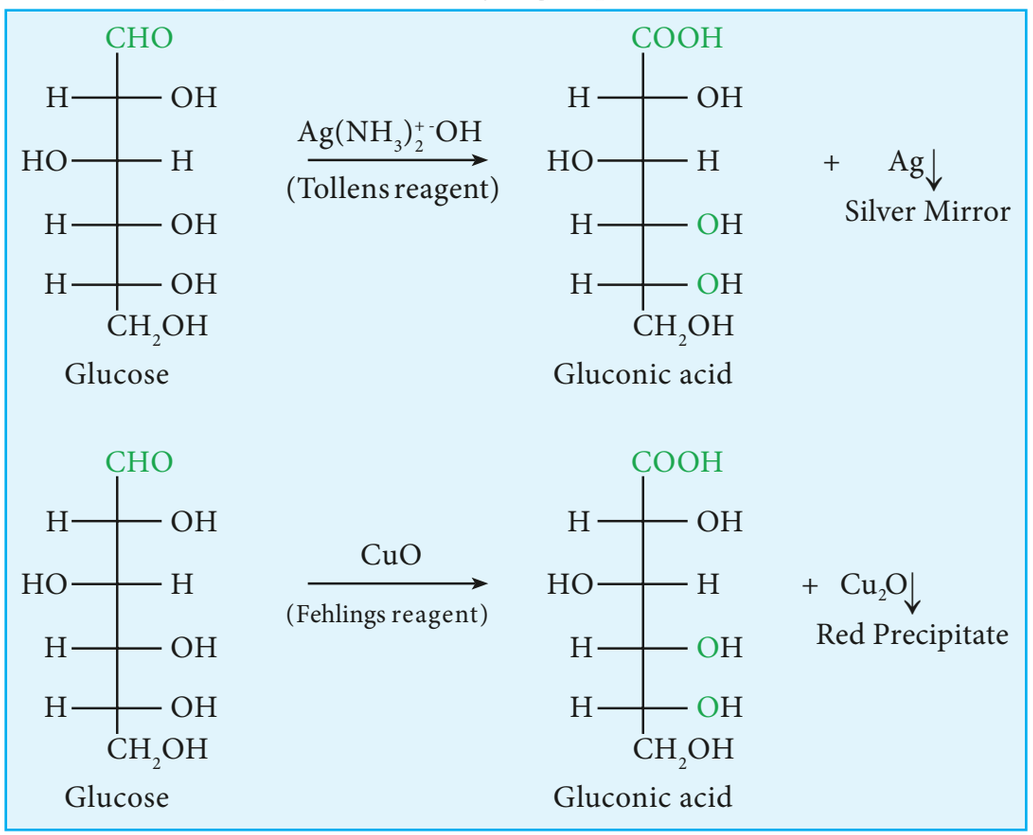
4. Glucose gets oxidized to gluconic acid with mild oxidizing agents like bromine water suggesting that the carbonyl group is an aldehyde group and it occupies one end of the carbon chain. When oxidised using strong oxidising agent such as conc. nitric acid gives glucaric acid (saccharic acid) suggesting the other end is occupied by a primary alcohol group.



5. Glucose is oxidised to gluconic acid with ammonical silver nitrate (Tollen's reagent) and alkaline copper sulphate (Fehling's solution). Tollen's reagent is reduced to metallic silver and Fehling's solution to cuprous oxide which appears as red precipitate. These reactions further confirm the presence of an aldehyde group



6. Glucose forms penta acetate with acetic anhydride suggesting the presence of five alcohol groups.

XII U14-Biomolecules_New.indd 242 11-01-2021 16:37:42